(Please write your Exam Roll No.) Exam Roll No. .....

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] JANUARY 2024

Paper Code: CIC-205 **Subject: Discrete Mathematics**

Time: 3 Hours **Maximum Marks: 75**

Note: Attempt five questions in all including Q. No.1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q1 Answer the following short answer type questions (Compulsory):-

(a) Define a power set and give an example. **(2)**
(b) State De Morgan's Laws in propositional logic. **(2)**
(c) What is the principle of mathematical induction? **(2)**
(d) Give an example of an equivalence relation. **(2)**
(e) Define a Boolean Algebra. **(2)**
(f) Explain the difference between a group and a monoid. **(2)**
(g) What is Euler's formula in graph theory? **(2)**
(h) Define a Hamiltonian circuit in a graph. **(1)**

UNIT-I

Q2 (a) Construct a Venn diagram for three sets A, B, C with the following conditions: $A \cap B \neq \emptyset, B \cap C \neq \emptyset, A \cap C \neq \emptyset$. **(5)**

(b) Create a truth table for the proposition $p \rightarrow (q \wedge \neg r)$ and determine its logical equivalence. **(5)**

(c) Define a binary relation R on the set {1, 2, 3, 4} where R contains (a, b) if and only if a divides b. Determine if R is reflexive, symmetric, and transitive, providing justification for each property. **(5)**

Q3 (a) Using the principle of inclusion and exclusion, find the number of integers between 1 and 100 that are divisible by 2, 3, or 5. Provide step-by-step calculations. **(5)**

(b) Prove that in a group of six people, at least three people either all know each other or none of them knows the others. Use the pigeonhole principle in your proof. **(5)**

(c) Given a set S of n elements, derive an expression for the number of ways to choose a non-empty subset of S. Validate your expression with n = 5. **(5)**

UNIT-II

Q4 (a) Let f(x) = 2x - 1 and g(x) = x/2 + 1. Find the compositions f(g(x)) and g(f(x)) and determine their domains. Illustrate each step and clarify the reasoning behind the determination of their domains. **(5)**

(b) Show that every finite distributive lattice can be represented as a lattice of sets, using the lattice formed by the subsets of the set {a, b, c} as an example. Provide a detailed explanation. **(5)**

(c) Design a logic circuit for the Boolean expression $\overline{(A+\overline{B})C} + \overline{AC}$ and simplify it using Karnaugh maps. Include a step-by-step process for the circuit and a detailed explanation of the Karnaugh map simplification. **(5)**

P.T.O.

CIC-205
P-1/2

(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

III rd Semester, November, 2022

Paper Code: CIC 205 **Subject: Discrete Mathematics**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1 (a)	Write the Principle of Inclusion and Exclusion ? Explain in brief.	2	CO1
(b)	Analyse Divide and conquer technique? Explain any one Example of it ?	2	CO2
(c)	Find the Converse and contrapositive of the Statement "If x is positive then $x \neq 0$"	2	CO1
(d)	Represent the statement using predicate and quantifier and negate it For all the real number x if $x > 5$ then $x^2 > 25$	2	CO1
(e)	Define lattices ?	2	CO2
2 (a)	Show that $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$ is tautology By Rules of Preposition.	5	CO1
(b)	Prove that "if x,y ∈ Z (set of integer)such that xy is odd then both x and y are odd, by proving its contrapositive	5	CO1
3 (a)	Minimize the given function using K-map $f(x,y,z) = xyz + \bar{x}yz + x\bar{y}z + xy\bar{z} + \overline{xyz}$	5	CO2
(b)	What are the different solution methods of homogenous and non homogenous recurrence relation Explain with example?	5	CO2
4 (a)	(a) Let A ={1,2,3,4,6} and R is a Relation on the Set A Such that aRb if a divides b Find (i) Relation R (ii) Diagraph of R (iii) Find Adjacency matrix of R (iv) Indergree and outdegree of each node (v) Find its Hasse diagram	5	CO1
(b)	Analyse all the three cases of master method in solving Recurrance Relation	5	CO2